

drcmd: Doubly-Robust Causal Inference with Missing Data

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Summary

`drcmd` is an R package for doubly-robust estimation of causal effects of binary treatments assigned at a single time point, with support for partially missing outcomes, treatments, and confounding covariates. The estimands include mean outcomes under treatment and control, average treatment effects, and causal risk and odds ratios for binary outcomes. Across these missing-data settings, the package combines inverse-probability weighting of complete cases with regressions on variables observed for all participants, through which incomplete cases also contribute to estimation. Users specify the methods used to fit these regressions, including machine learning methods, and can enable cross-fitting to separate regression fitting from treatment-effect estimation. Point estimates are accompanied by standard errors and confidence intervals, as well as a description of the missing-at-random condition required for the supplied data. The plausibility of this condition must be assessed together with the causal identification assumptions for the study.

Statement of need

Missing-data problems in causal inference differ in both the variables that are unavailable and the information that remains available to account for their missingness. An electronic health record study, for example, may have complete treatment and covariate information but incomplete outcome ascertainment. In a two-phase validation study, accurate outcome and treatment measurements may instead be obtained for a subset of patients selected on the basis of covariates and error-prone measurements recorded for everyone. The gold-standard measurements are then missing outside the validation sample, while the error-prone measurements can be used to account for selection into that sample and to predict the unavailable measurements. Here, the sampling probabilities are often known by design, whereas probabilities of outcome ascertainment in the first example would generally need to be estimated.

Although the causal estimand may be the same in these studies, an estimator developed for one observed-data structure cannot generally be applied to the other without modification. When treatment or covariate information is partially missing, unweighted complete-case regressions need not estimate the outcome regression and treatment propensity score in the target population, because their fitting sample has been selected according to the availability of the analysis variables. Nuisance fitting must account for this selection, while an augmentation based on variables that remain observed can use information from incomplete cases.

`drcmd` implements these adjustments for researchers estimating point-treatment effects from incomplete observational and two-phase data. We use the relationship between complete-data and observed-data influence functions (Robins et al. 1994; Tsiatis 2006) to construct the estimator from the variables that are always observed and those that are partially missing. Researchers can therefore use the same fitting procedure when the set of incomplete variables changes, while specifying regression learners and, where available, known sampling probabilities and auxiliary measurements. In the validation example, these auxiliary variables are the error-prone measurements, which inform estimation of effects defined in terms of the gold-standard variables.

State of the field

Several R packages implement doubly-robust estimators with support for specific missing-data settings. `AIPW` (Zhong et al. 2021) and `tmle` (Gruber and van der Laan 2012) accommodate missing outcomes with fully observed treatments and covariates. `drtmle` additionally permits missing treatments, while retaining fully observed covariates (Benkeser et al. 2017). `twoStageDesignTMLE` implements targeted maximum likelihood estimation when some covariates are collected through two-phase sampling (Rose and van der Laan 2011).

`SMMAL` implements semi-supervised estimation when treatment and outcome labels are jointly missing but covariates and surrogate measurements remain observed (Hou et al. 2021). Missing covariates can also be accommodated through the forest splitting procedure in `grf` (Athey et al. 2019), or by using `MatchThem` to perform matching or weighting across multiply imputed datasets (Pishgar et al. 2021).

Extending a package organized around missing outcomes to support partially observed treatments and covariates would require changes to its nuisance fitting, estimator construction, and variance calculations. We implemented these parts of `drcmd` around the partition into always-observed and partially observed variables, with support for known sampling probabilities and empirical efficiency maximization under the assumptions described below. The regression algorithms themselves are provided by `SuperLearner`, whose learner libraries and ensemble fitting routines are used for each nuisance function.

Software design

Let $F = (Y, A, X)$ denote the full-data observation that would be available in the absence of missingness, where Y is the outcome, A is a binary treatment, and X contains the covariates required for confounding adjustment. Users supply these variables with missing values recorded as `NA`, and may additionally supply fully observed auxiliary variables W that predict the partially observed variables or their missingness but would not be included in the corresponding complete-data causal analysis. In two-phase validation studies, for example, W may contain error-prone measurements of variables whose gold-standard values are available only in the validation sample.

`drcmd` collects all always-observed variables in Z , all partially observed variables in U , and constructs a complete-case indicator R that equals one when every variable in F is observed. The package requires at least one always-observed variable and some complete cases, although these requirements alone do not ensure identification of the causal estimands. In addition to consistency, conditional exchangeability given X , and treatment positivity, identification requires the

missing-at-random condition and complete-case positivity condition

$$U \perp R \mid Z, \quad P(R = 1 \mid Z) > 0.$$

The fitted output identifies the variables comprising U and Z and reports the implied conditional independence assumption, whose plausibility must be assessed for the particular application.

The complete-case representation discards measurements that may be available under distinct incomplete-data patterns: if any variable in U is missing for a subject, observed values of the remaining variables in U are not used for that subject, whose contribution is based on Z and R . Consequently, the package may sacrifice efficiency relative to an estimator that uses these additional measurements, and the condition $U \perp R \mid Z$ is not implied by a general missing-at-random assumption for the original missingness patterns.

For treatment level a , let $\chi_a(F)$ denote the mean-zero complete-data efficient influence function for the counterfactual mean $\psi_a = E\{Y(a)\}$. Under the stated assumptions, the influence function for the reduced observation $O = (Z, R, RU)$ is (Robins et al. 1994; Tsiatis 2006)

$$D_a(O) = \frac{R}{\kappa(Z)} \chi_a(F) - \left\{ \frac{R}{\kappa(Z)} - 1 \right\} \varphi_a(Z),$$

where $\kappa(Z) = P(R = 1 \mid Z)$ and $\varphi_a(Z) = E\{\chi_a(F) \mid Z, R = 1\}$. The regression $\varphi_a(Z)$ augments the inverse-probability-weighted complete-case contribution by estimating the conditional expectation of $\chi_a(F)$ given variables that are also available for incomplete cases.

Together with $\kappa(Z)$ and $\varphi_a(Z)$, estimation requires the complete-data outcome regression $m_a(X) = E(Y \mid A = a, X)$ and treatment propensity score $g_a(X) = P(A = a \mid X)$ that enter $\chi_a(F)$. The latter two functions are estimated using losses weighted by $R/\hat{\kappa}(Z)$ to account for selection into the complete cases, with known complete-case probabilities used in place of $\hat{\kappa}(Z)$ when supplied by the user. All four nuisance functions are fit through **SuperLearner** (van der Laan et al. 2007), using a common learner library or separately specified libraries. The use of weighted losses restricts this choice to learners that respect observation weights, since learners that ignore them do not minimize the specified losses and can invalidate the resulting estimates.

With these nuisance estimates, **drcmd** adds the empirical mean of the estimated influence function to an initial plug-in estimate to obtain its default one-step estimator. Targeted maximum likelihood estimation is also available and is asymptotically equivalent under suitable regularity conditions. Both can be implemented with cross-fitting, in which the nuisance functions are fit on training observations and the correction or targeting step is evaluated on held-out observations. This relaxes empirical-process restrictions on the nuisance estimators, although their convergence rates must still be sufficient for valid inference. Since cross-fitting repeats the nuisance fits for each training sample, folds can be run in parallel on supported operating systems to reduce computation time.

For the pseudo-outcome regression, users can replace the usual prediction loss with an empirical second moment of the estimated observed-data influence function. This empirical efficiency maximization criterion targets the variance of the treatment-effect estimator, and was included to address the sensitivity of that variance to estimation of φ_a when relatively few complete cases are available (Barnatchez et al. 2025).

Counterfactual means and their contrasts are reported with influence-function-based standard errors and confidence intervals, with effects among treated or control subjects additionally available under one-step estimation. Diagnostic plots display the distributions of estimated treatment and

complete-case probabilities, estimated influence-function values, and pseudo-outcome regression fits for examining instability in the weighting and augmentation terms. The executable vignette documents the construction of each estimator and illustrates learner specification and interpretation of the fitted output.

Research impact statement

`drcmd` was initially developed for our work on causal effect estimation under two-phase sampling with error-prone outcome and treatment measurements, where it was used to implement the complete-data-based one-step estimators evaluated in the simulation study, including variants based on empirical efficiency maximization (Barnatchez et al. 2025). The methods were also evaluated using Vanderbilt Comprehensive Care Clinic records to estimate the effect of initiating antiretroviral therapy within one month of entry to care on the three-year risk of an AIDS-defining event. In that application, gold-standard treatment and outcome measurements were sampled for validation with probabilities that depended on covariates and error-prone measurements, so that estimation had to account for differences between the validation sample and the full cohort.

The implementation is available in the package repository, together with automated tests covering estimation, missing-data handling, and cross-fitting, and continuous-integration checks on Linux, macOS, and Windows. The package documentation includes an executable vignette and function references for applying the estimators beyond the jointly missing outcome and treatment setting considered in the methods study.

AI usage disclosure

Generative AI assisted with the initial manuscript draft. OpenAI Codex (GPT-6) assisted with revising the prose, checking software descriptions against the source code and documentation, and checking references and JOSS formatting requirements.

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